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Yu et al.

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(54) **INTEGRATED FAN-OUT PACKAGES AND METHODS OF FORMING THE SAME**

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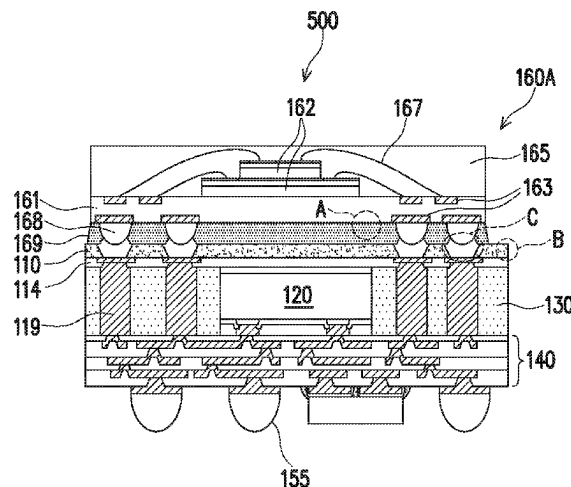
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(57) **ABSTRACT**

A method includes forming a composite material layer over a carrier, the composite material layer including particles of a filler material incorporated into a base material, forming a set of through vias over a first side of the composite material layer, attaching a die over the first side of the composite material layer, the die being spaced apart from the set of through vias, forming a molding material over the first side of the composite material layer, the molding material least laterally encapsulating the die and the through vias of the set of through vias, forming a redistribution structure over the die and the molding material, the redistribution structure electrically connected to the through vias, forming openings in a second side of the composite material layer opposite the first side, and forming conductive connectors in the openings, the conductive connectors electrically connected to the through vias.

20 Claims, 13 Drawing Sheets



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(2013.01)

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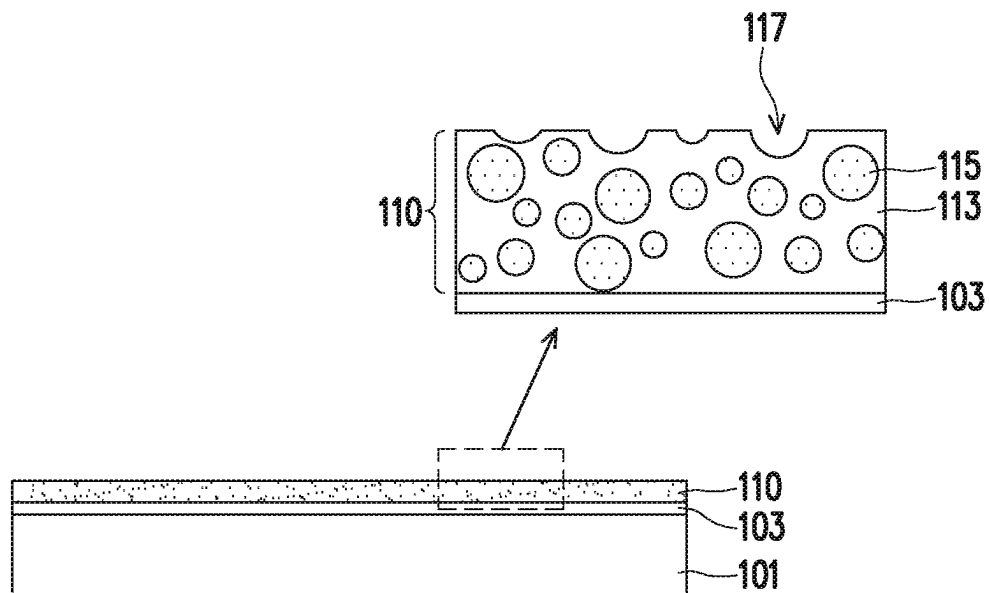


FIG. 1

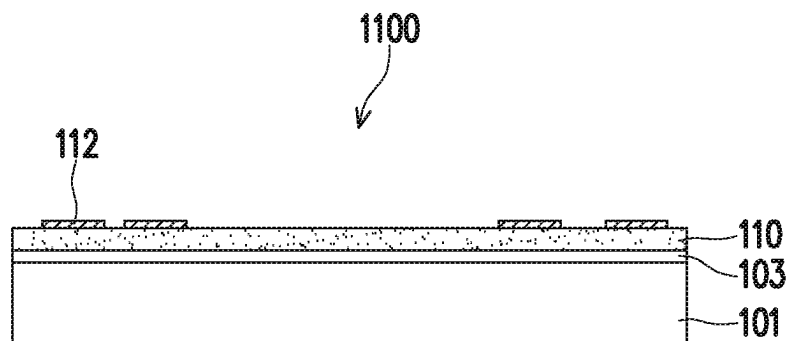


FIG. 2

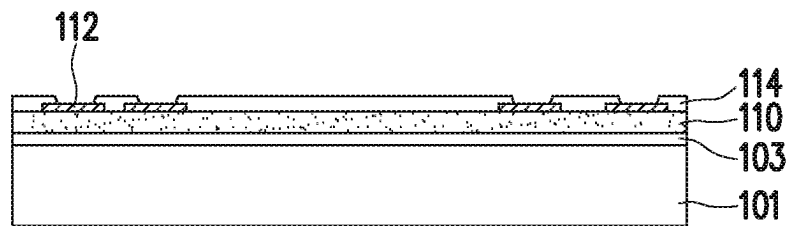


FIG. 3

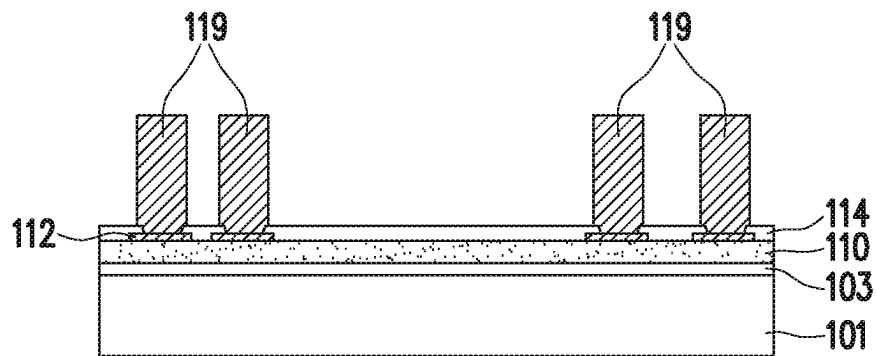


FIG. 4

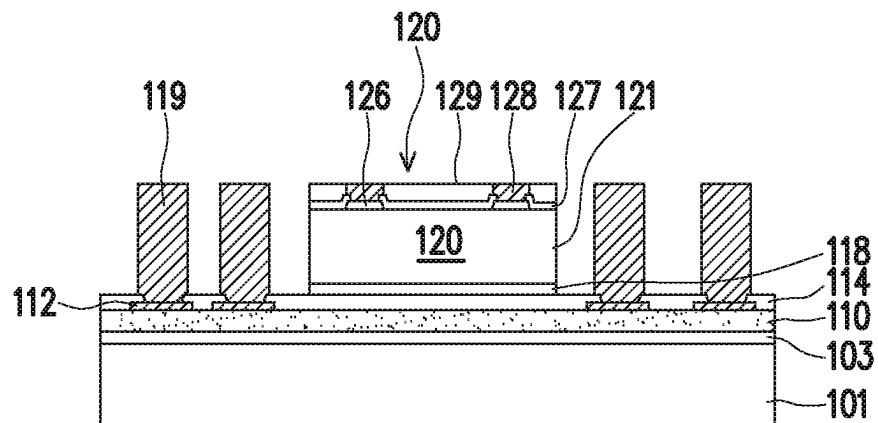


FIG. 5

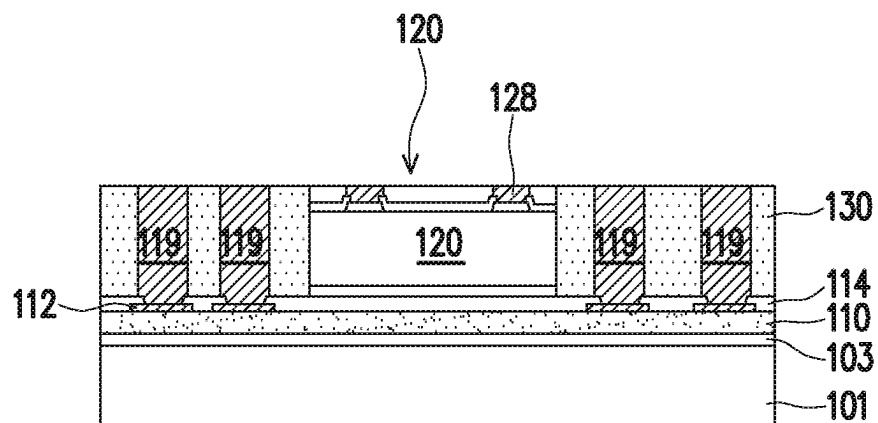


FIG. 6

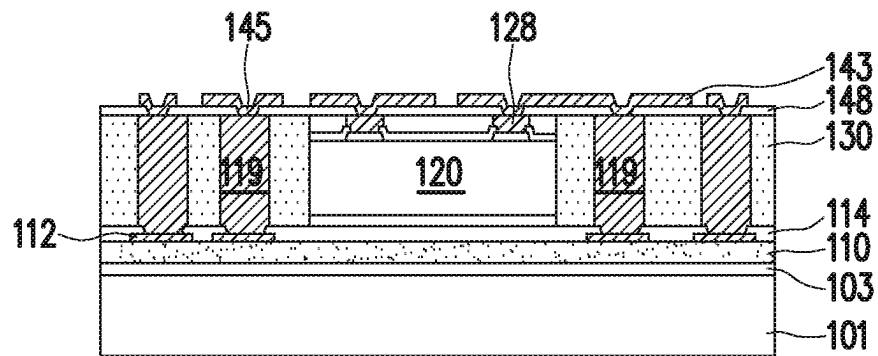


FIG. 7

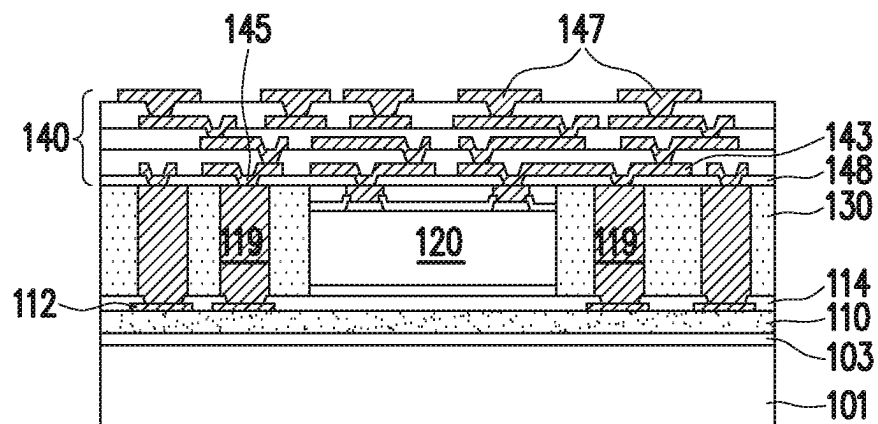


FIG. 8

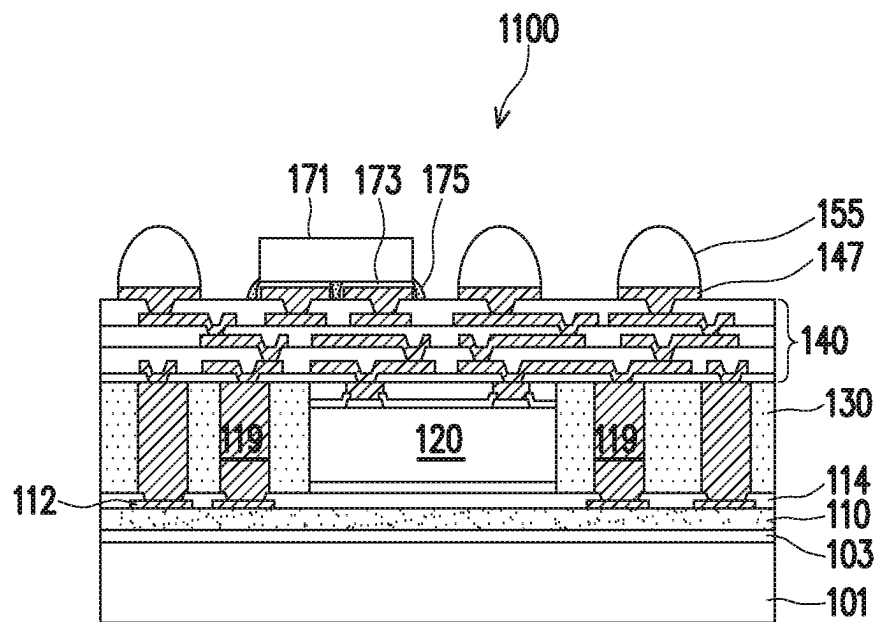


FIG. 9

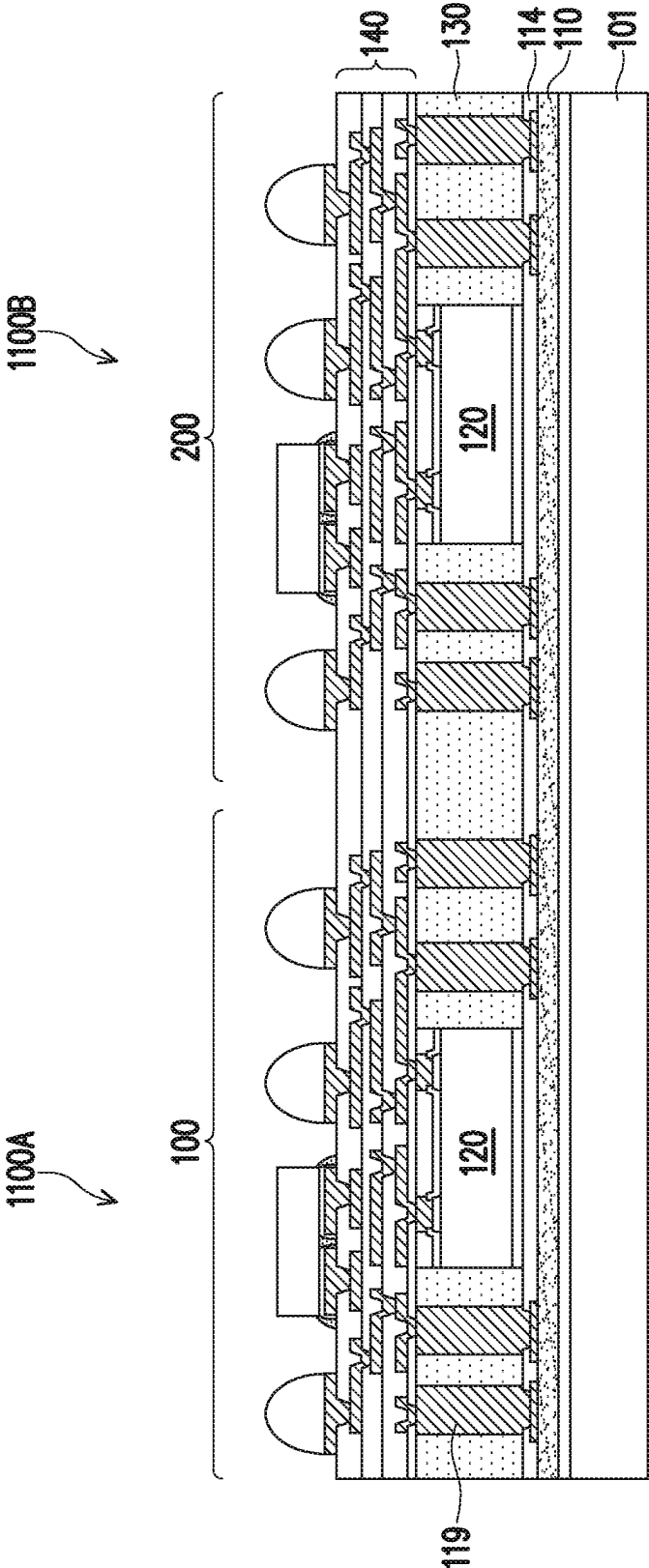


FIG. 10

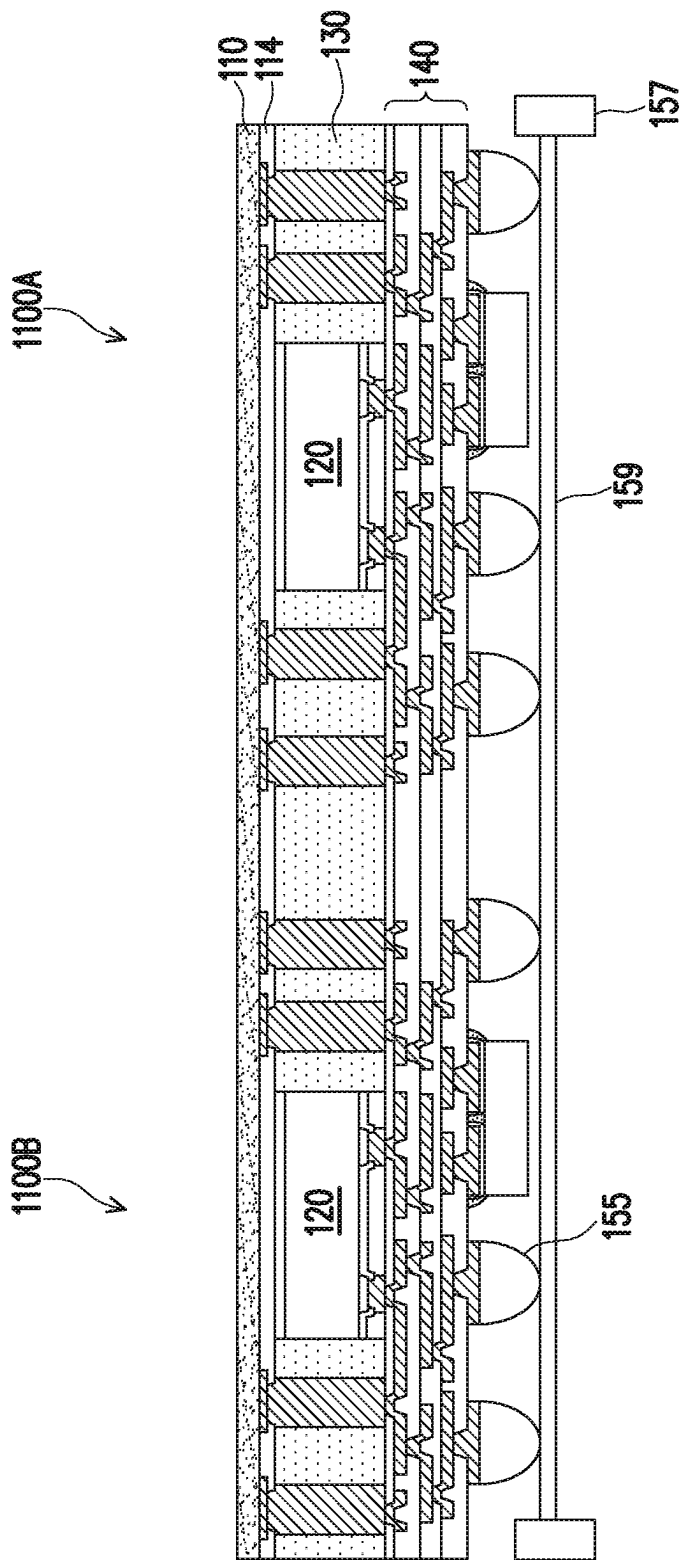


FIG. 11

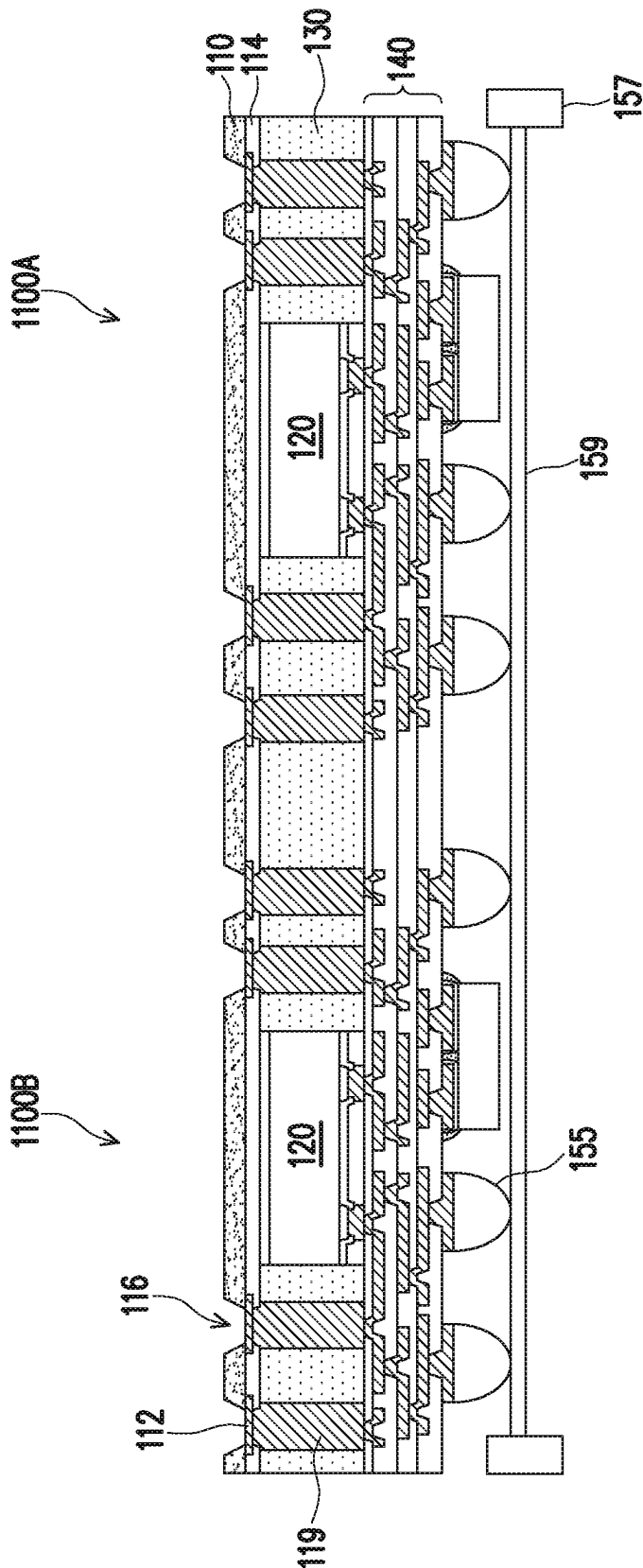


FIG. 12

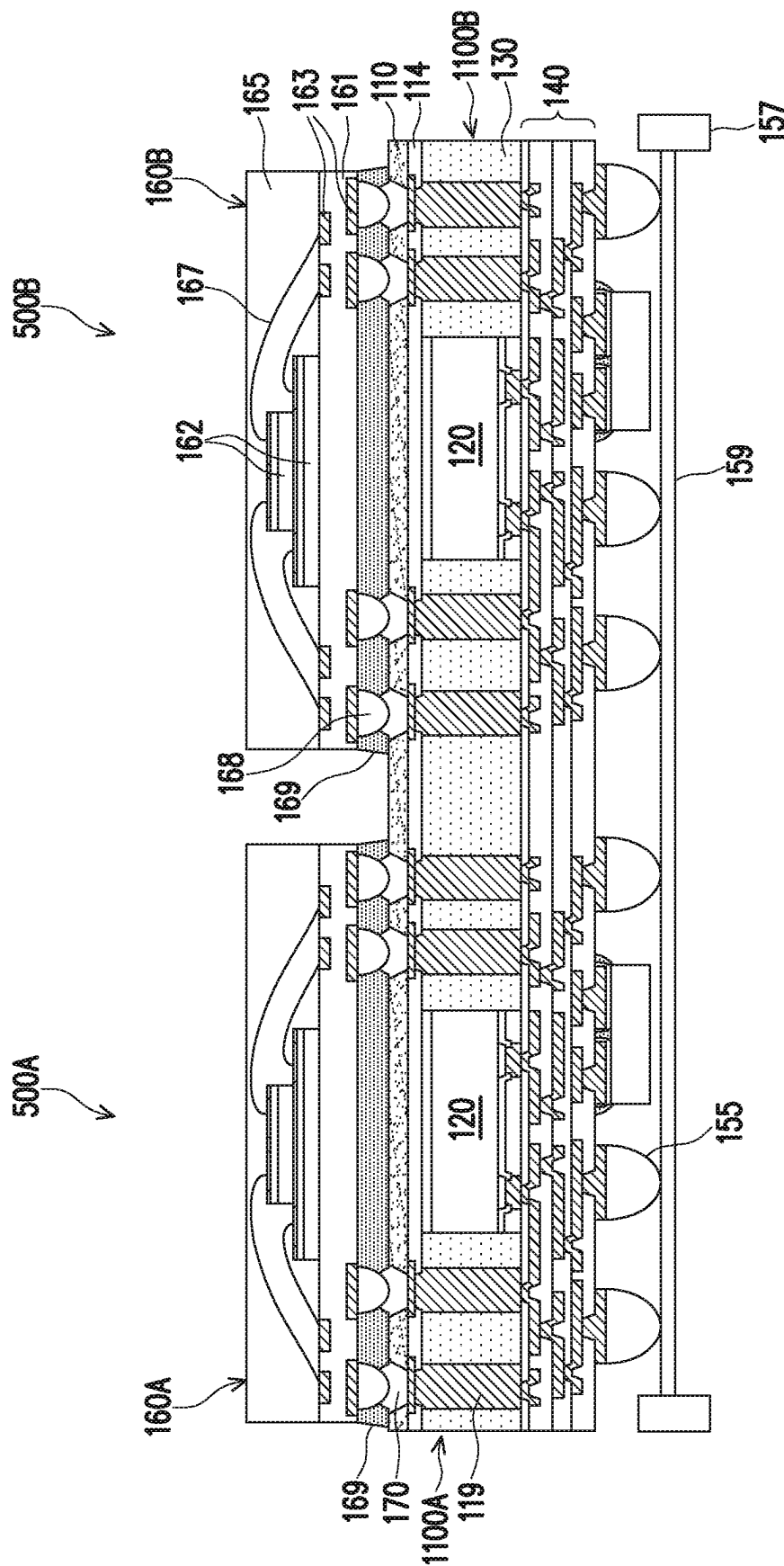


FIG. 13

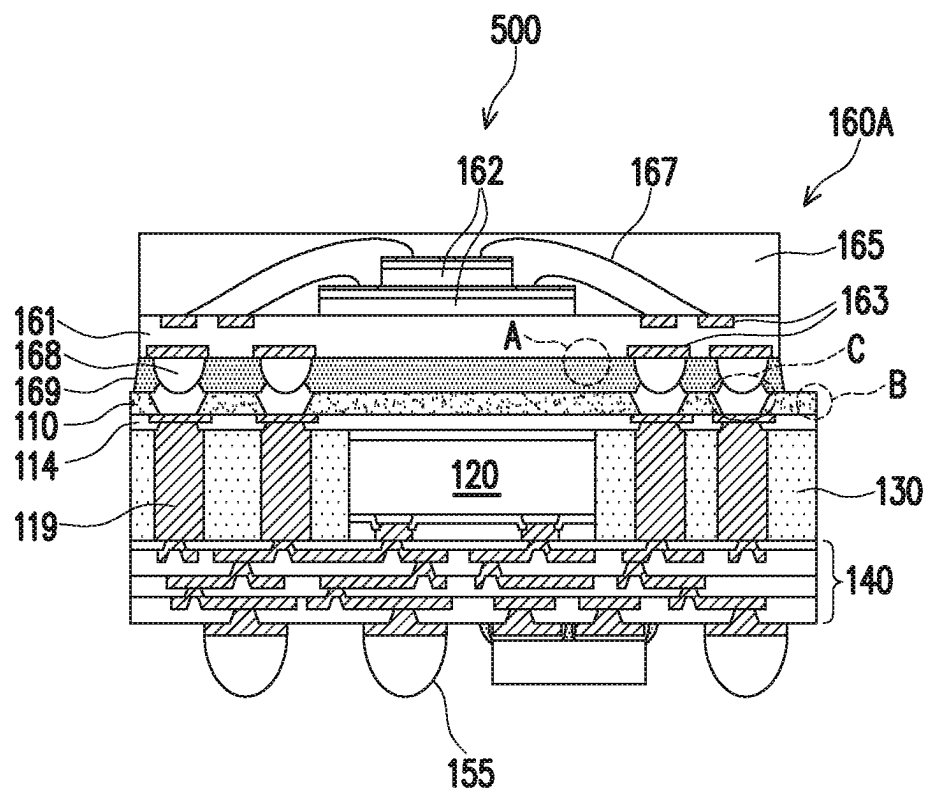


FIG. 14

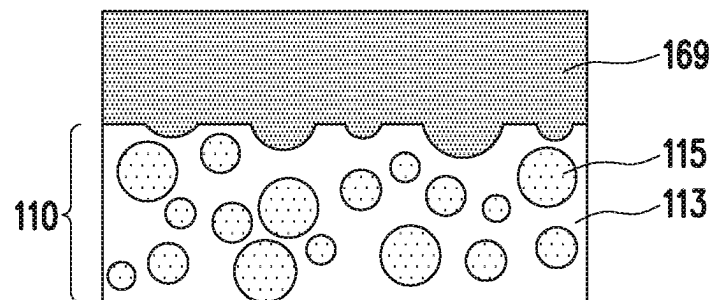


FIG. 15A

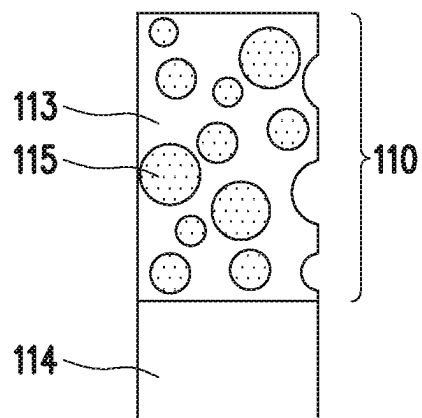


FIG. 15B

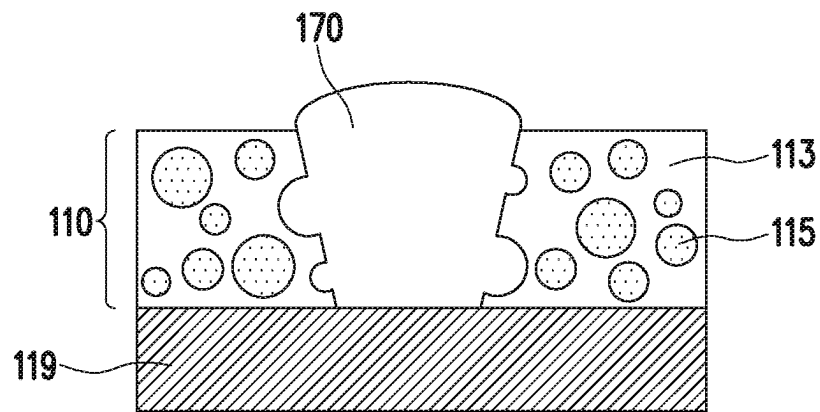


FIG. 15C

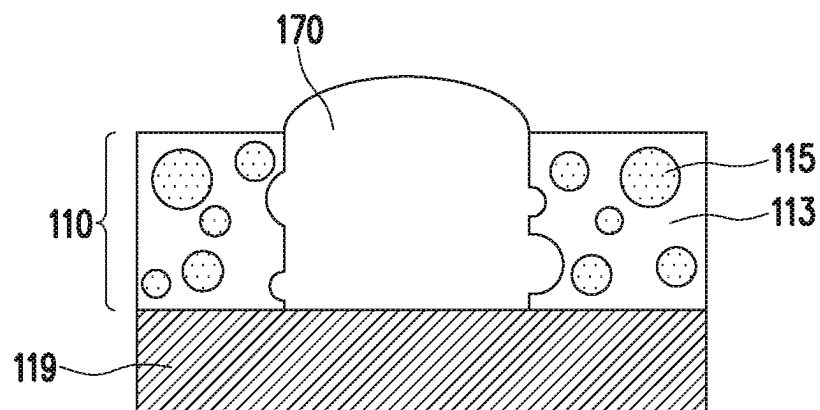


FIG. 15D

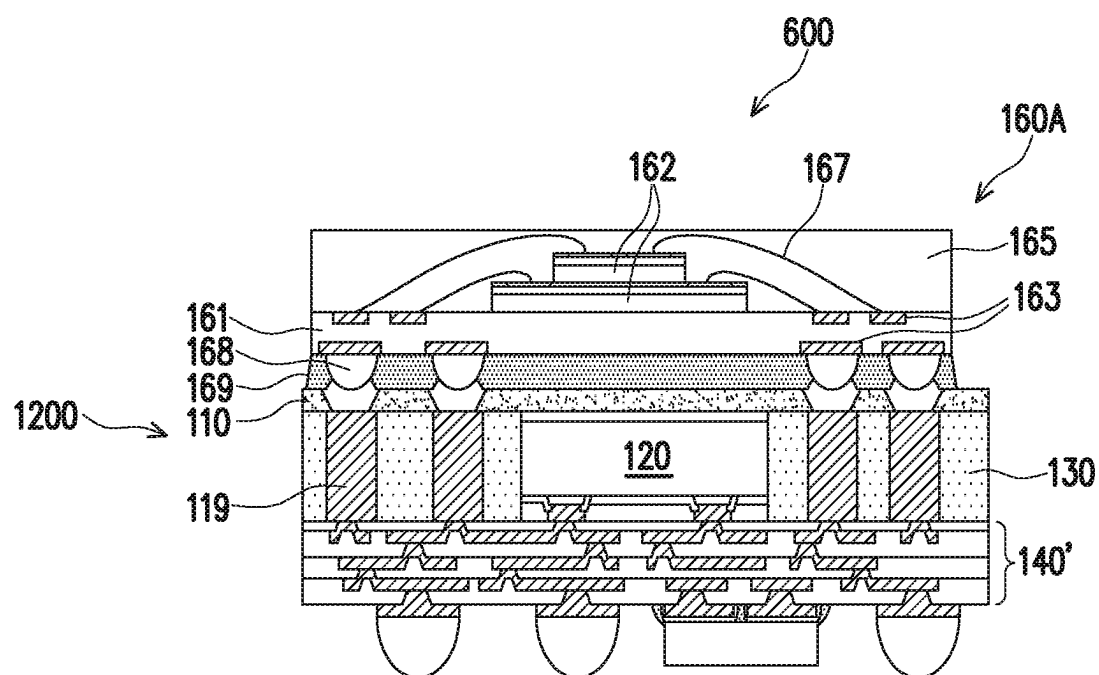


FIG. 16

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INTEGRATED FAN-OUT PACKAGES AND METHODS OF FORMING THE SAME

PRIORITY CLAIM AND CROSS-REFERENCE

This application is a continuation application of U.S. patent application Ser. No. 16/529,989, filed on Aug. 2, 2019, which claims benefit of U.S. Provisional Patent Application No. 62/727,311 filed Sep. 5, 2018, entitled “InFO Structure for Package on Package Devices and Methods of Forming the Same,” which applications are incorporated herein by reference.

BACKGROUND

The semiconductor industry has experienced rapid growth due to continuous improvements in the integration density of a variety of electronic components (e.g., transistors, diodes, resistors, capacitors, etc.). For the most part, this improvement in integration density has come from repeated reductions in minimum feature size, which allows more components to be integrated into a given area. As the demand for even smaller electronic devices has grown recently, there has grown a need for smaller and more creative packaging techniques of semiconductor dies.

An example of these packaging technologies is the Package-on-Package (POP) technology. In a PoP package, a top semiconductor packages is stacked on top of a bottom semiconductor package to allow high level of integration and component density. Another example is the Multi-Chip-Module (MCM) technology, where multiple semiconductor dies are packaged in one semiconductor package to provide semiconductor devices with integrated functionalities.

The high level of integration of advanced packaging technologies enables production of semiconductor devices with enhanced functionalities and small footprints, which is advantageous for small form factor devices such as mobile phones, tablets and digital music players. Another advantage is the shortened length of the conductive paths connecting the interoperating parts within the semiconductor package. This improves the electrical performance of the semiconductor device, since shorter routing of interconnections between circuits yields faster signal propagation and reduced noise and cross-talk.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 illustrates a cross-sectional view of a composite layer in a semiconductor package at a stage of fabrication, in accordance with an embodiment.

FIGS. 2-13 illustrate cross-sectional views of a semiconductor package at various stages of fabrication, in accordance with an embodiment.

FIG. 14 illustrates a cross-sectional view of a semiconductor package, in accordance with an embodiment.

FIGS. 15A-D illustrate various views of composite layers in a semiconductor package at various stages of fabrication, in accordance with some embodiments.

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FIG. 16 illustrates a cross-sectional view of a semiconductor package, in accordance with an embodiment.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Embodiments of the present disclosure are discussed in the context of semiconductor packages and methods of forming the semiconductor packages, and in particular, integrated fan-out (InFO) semiconductor packages. A layer of a composite material that includes a filler material (e.g., particles) incorporated into a dielectric material (e.g., a polymer) is formed over a carrier, and then one or more semiconductor dies and/or conductive pillars are formed over the composite material. A molding material is formed over the carrier and around the dies and around the conductive pillars. A redistribution structure is formed over the molding material, the dies and the conductive pillars. In some cases, the use of a composite material layer can improve the structural rigidity of the semiconductor package. The layer of the composite material can also reduce warping or bending due to other layers such as those of the redistribution structure. Additionally, the layer of the composite material may have a rough or pitted surface which can improve adhesion of material subsequently deposited on the composite material.

FIG. 1 illustrates a cross-sectional view of a composite layer 110 in a package structure 500 at a stage of fabrication, in accordance with an embodiment. FIGS. 2-13 illustrate cross-sectional views of the package structure 500 at various stages of fabrication, in accordance with an embodiment. FIG. 14 illustrates a cross-sectional view of the package structure 500, in accordance with an embodiment. FIGS. 15A-C illustrate various views of composite layers 100 in a package structure 500 at various stages of fabrication, in accordance with some embodiments. FIG. 16 illustrates a cross-sectional view of a package structure 600, in accordance with an embodiment.

Referring to FIG. 1, a release layer 103 and a composite layer 110 are formed over a carrier 101. The carrier 101 may be a wafer, a panel structure, or the like, and may be made of a material such as silicon, silicon oxide, aluminum, aluminum oxide, polymer, polymer composite, metal foil, ceramic, glass, glass epoxy, beryllium oxide, tape, the like,

or a combination. The carrier **101** provides support for subsequently formed structures.

In some embodiments, a release layer **103** is deposited or laminated over the carrier **101** before the composite layer **110** is formed. The release layer **103** may be formed of a polymer-based material, which may be removed along with the carrier **101** from overlying structures formed in subsequent steps. In some embodiments, the release layer **103** is an epoxy-based thermal-release material, which loses its adhesive property when heated, such as a Light-to-Heat Conversion (LTHC) release coating. In other embodiments, the release layer **103** may be a photosensitive material such as an ultra-violet (UV) glue, which loses its adhesive property when exposed to UV light. The release layer **103** may be dispensed as a liquid and cured, may be a laminate film laminated onto the carrier **101**, or the like. The top surface of the release layer **103** may be leveled and may have a high degree of co-planarity.

Still referring to FIG. 1, the composite layer **110** is formed over the release layer **103**. FIG. 1 also illustrates a magnified portion of the composite layer **110**. In some embodiments, the composite layer **110** is a composite material including a filler material **115** incorporated within a base material **113**. The filler material **115** may increase the mechanical strength or rigidity of the composite layer **110**, described in greater detail below. The base material **113** may be a polymer, epoxy, resin, underfill material, a combination of materials, or the like.

The filler material **115** of the composite layer **110** may comprise particles, fibers, the like, or a combination. In some embodiments, the filler material **115** comprises particles of silicon oxide, aluminum oxide, the like, or a combination. In some embodiments, the particles have a diameter between about 0.5 μm and about 30 μm , though the particles may have other diameters in other embodiments. In some embodiments, the filler material **115** of the composite layer **110** may be selected to have a particular range of diameters or to have an average diameter. For example, in some embodiments, the filler material **115** may be selected to have an average diameter of between about 0.5 μm and about 30 μm . In some embodiments, the volume of filler material **115** within the composite layer **110** may be between about 30% and about 80% of the total volume of the composite layer **110**. In some embodiments, the volume ratio of filler material **115** to base material **113** may be between about 0.5:1 and about 3:1. The characteristics of the filler material **115** may be selected to provide a particular characteristic to the composite layer **110**, such as rigidity. For example, a composite layer **110** with filler material **115** of a larger average diameter may have more rigidity (e.g., a larger Young's modulus) than a composite layer **110** with filler material of a smaller average diameter. By using a material for the composite layer **110** that has a greater rigidity, the rigidity of the structure formed thereon (e.g., package structure **500** in FIG. 14) may be improved, and warping or bending of the structure may be reduced (described in greater detail below).

In some embodiments, the composite layer **110** is a composite polymer material, an underfill material, a molding compound, an epoxy, a resin, a combination of materials, or the like. In some embodiments, the composite layer **110** may have a coefficient of thermal expansion (CTE) that is greater than about 10 $\text{ppm}/^\circ\text{C}$., such as about 22 $\text{ppm}/^\circ\text{C}$. In some embodiments, the composite layer **110** may have a Young's modulus greater than about 10 GPa, such as about 23 GPa. In some embodiments, the composite layer **110** may have a thickness between about 10 μm and about 100 μm , such as

about 35 μm . The composite layer **110** may be formed over the carrier **101** using a suitable deposition process, such as spin coating, chemical vapor deposition (CVD), laminating, the like, or a combination thereof. In some embodiments, the composite layer **110** is cured after deposition using a curing process. The curing process may comprise heating the composite layer **110** to a predetermined temperature for a predetermined period of time, using an anneal process or other heating process. The curing process may also comprise an ultra-violet (UV) light exposure process, an infrared (IR) energy exposure process, combinations thereof, or a combination thereof with a heating process. Alternatively, the composite layer **110** may be cured using other techniques. In some embodiments, a curing process is not included.

In some cases, one or more surfaces of the composite layer **110** may be pitted, and as such include pits **117**, as shown in FIG. 1. The pits **117** may be caused by, for example, exposed pieces of the filler material **115** becoming dislodged or otherwise removed from the base material **113**, leaving behind pits **117** where the pieces of filler material **115** had been previously located. For example, exposed pieces of the filler material **115** may become dislodged during a subsequent cleaning process or during another subsequent process step. In some cases, some of the pits **117** may have a size (e.g., diameter or depth) approximately equal to or less than the size (e.g., diameter) of the filler material **115**. For example, some of the pits **117** may have a diameter or depth between about 0.5 μm and about 30 μm , in some embodiments. However, in some cases some pits **117** may have a size less than the size of the filler material **115** or a size greater than that of the filler material **115**. In some cases, the presence of the pits **117** may improve adhesion of overlying layers, such as the dielectric layer **114** shown in FIG. 3. In some cases, the presence of pits **117** may cause a surface of the composite layer **110** to have a roughness between about 0.1 μm and about 10 μm . In some cases, the pits **117** may cover between about 50% and about 90% of a surface of the composite layer **110**.

Turning to FIG. 2, a metallization pattern **112** is formed on the composite layer **110**. In some embodiments, the metallization pattern **112** is formed by forming a seed layer (not shown) over the composite layer **110**. The seed layer may be a metal layer or another type of layer, and may include one or more layers of one or more different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photoresist is then formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the metallization pattern **112**. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, a combination, or the like. Then, the photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. In some embodiments, once the photoresist is removed, exposed portions of the seed layer are removed using an etching process, such as a wet etching process or a dry etching process. The remaining portions of the seed layer and conductive material form the metallization pattern **112**.

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FIG. 3, a dielectric layer **114** is formed on the metallization pattern **112** and the composite layer **110**. In some embodiments, the dielectric layer **114** is formed of a polymer, which may be a photo-sensitive material such as PBO, polyimide, BCB, or the like, that may be patterned using a lithography mask. In other embodiments, the dielectric layer **114** is formed of a nitride such as silicon nitride, an oxide such as silicon oxide, PSG, BSG, BPSG, or the like. The dielectric layer **114** may be formed by spin coating, lamination, CVD, the like, or a combination thereof. The dielectric layer **114** is patterned to expose portions of the metallization pattern **112**. The dielectric layer **114** may be patterned using an acceptable process, such as by exposing the dielectric layer **114** to light when the dielectric layer **114** is a photo-sensitive material. In some embodiments, the dielectric layer **114** may be patterned using an etching mask and a suitable etching process such as an anisotropic etching process. In some embodiments, additional metallization patterns and dielectric layers may be formed in a stack over the metallization pattern **112** and dielectric layer **114**, using similar techniques.

Turning to FIG. 4, through vias **119** are formed over the metallization pattern **112** and the dielectric layer **114**. In some embodiments, the through vias **119** may be formed by forming a seed layer over the dielectric layer **114** and then forming a patterned photoresist over the seed layer, where each of the openings in the patterned photoresist corresponds to a location of the through via **119** to be formed. The openings in the dielectric layer **114** are filled with an electrically conductive material such as copper using a suitable technique such as electroplating or electroless plating. The photoresist is then removed using a suitable process such as an ashing or a stripping process. Portions of the seed layer on which the through vias **119** are not formed may then be removed using a suitable etching process. The through vias **119** may be formed as conductive pillars extending above the metallization pattern **112** and the dielectric layer **114**. Other techniques for forming the through vias **119** are also possible and are fully intended to be included within the scope of the present disclosure.

Next, in FIG. 5, a semiconductor die **120** (may also be referred to a die, or an integrated circuit (IC) die) is attached to the upper surface of the dielectric layer **114**. An adhesive film **118**, such as a die attach film (DAF), may be used to attach the die **120** to the dielectric layer **114**. The die **120** may be attached using a suitable process such as a pick-and-place process. In some embodiments, the DAF may be cured after the die **120** is attached.

Before being adhered to the dielectric layer **114**, the die **120** may be processed according to applicable manufacturing processes to form integrated circuits in the die **120**. For example, the die **120** may include a semiconductor substrate and one or more overlying metallization layers, collectively illustrated in FIG. 5 as element **121**. The semiconductor substrate may be, for example, doped or undoped silicon, or an active layer of a semiconductor-on-insulator (SOI) substrate. The semiconductor substrate may include other semiconductor materials such as germanium, a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, gallium nitride, indium phosphide, indium arsenide, and/or indium antimonide, an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP, combinations thereof, or the like. Other substrates such as multi-layered or gradient substrates may also be used. The die **120** may include devices (not shown), such as transistors, diodes, capacitors, resistors, etc., that are formed in and/or on the semiconduc-

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tor substrate and may be interconnected by the metallization layers to form an integrated circuit. The metallization layers may include metallization patterns in one or more dielectric layers over the semiconductor substrate (e.g., as a redistribution structure).

The die **120** further comprises pads **126** (e.g., contact pads, aluminum pads, or the like) to which external connections may be made. The pads **126** may be located on the front side (e.g., the “active side”) of the die **120**. A passivation film **127** may be formed over the front side of the die **120** and on portions of the pads **126**. Openings may be formed extending through the passivation film **127** to the pads **126**. Die connectors **128** extend into the openings of the passivation film **127** and are mechanically and electrically coupled to the respective pads **126**. The die connectors **128** may be, for example, conductive pads or conductive pillars. The die connectors **128** may comprise one or more conductive materials such as copper, and may be formed using a suitable process such as plating. The die connectors **128** are electrically coupled to devices and/or integrated circuits of the die **120**.

A dielectric material **129** may be formed at the active sides of the die **120**, such as on the passivation film **127** and/or the die connectors **128**. The dielectric material **129** laterally encapsulates the die connectors **128**, and the dielectric material **129** is laterally coterminous with the die **120**. The dielectric material **129** may be a polymer such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), a nitride such as silicon nitride or the like, an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), a combination, or the like. The dielectric material **129** may be formed, for example, by spin coating, lamination, CVD, or the like.

Next, in FIG. 6, a molding material **130** is formed over the dielectric layer **114**. The molding material laterally surrounds the die **120** and laterally surrounds the through vias **119**, separating the through vias **119** from the die **120** and from each other. The molding material **130** may comprise an epoxy, an organic polymer, a polymer with or without a silica-based or glass filler added, or other materials, as examples. In some embodiments, the molding material **130** comprises a liquid molding compound (LMC) that is a gel type liquid when applied. The molding material **130** may also comprise a liquid or solid when applied. Alternatively, the molding material **130** may comprise other insulating or encapsulating materials. The molding material **130** is applied using a wafer level molding process in some embodiments. The molding material **130** may be molded using, for example, compressive molding, transfer molding, or other techniques.

In some embodiments, the molding material **130** may be cured using a curing process. The curing process may comprise heating the molding material **130** to a predetermined temperature for a predetermined period of time, using an anneal process or other heating process. The curing process may also comprise an ultra-violet (UV) light exposure process, an infrared (IR) energy exposure process, combinations thereof, or the like. Alternatively, the molding material **130** may be cured using other techniques. In some embodiments, a curing process is not preformed.

Still referring to FIG. 6, a planarization process, such as chemical-mechanical polish (CMP), may optionally be performed to remove excess portions of the molding material **130** over the front side of the die **120**. After the planarization

process, the molding material **130**, the through vias **119**, and the die connectors **128** may have top surfaces that are coplanar.

Referring next to FIG. 7 and FIG. 8, a redistribution structure **140** is formed over the molding material **130**, the through vias **119**, and the front side of the die **120**, in accordance with some embodiments. The redistribution structure **140** comprises one or more layers of electrically conductive features (e.g., metallization patterns including conductive lines **143**, vias **145**, and the like) formed in one or more dielectric layer (e.g., dielectric layer **148**).

In some embodiments, the one or more dielectric layers (e.g., dielectric layer **148**) are formed of a polymer, such as polybenzoxazole (PBO), polyimide, benzocyclobutene (BCB), a photo-sensitive polymer, or the like. In some embodiments, the one or more of the dielectric layers may include other materials such as a nitride (e.g., silicon nitride), an oxide (e.g., silicon oxide), phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like. The one or more dielectric layers may be formed by a suitable deposition process, such as spin coating, chemical vapor deposition (CVD), laminating, the like, or a combination thereof.

In FIG. 7, the dielectric layer **148** is formed over the molding material **130**, the through vias **119**, and the front side of the die **120**, and then patterned. The patterning forms openings to expose portions of the through vias **119** and the die connectors **128** of the die **120**. The dielectric layer **148** may be patterned using an acceptable process, such as by exposing the dielectric layer **148** to light when the dielectric layer **148** is a photo-sensitive material and developing the dielectric layer **148** after the exposure to form the openings. The dielectric layer **148** may also be patterned by etching using, for example, an anisotropic etch.

Still referring to FIG. 7, a metallization pattern including conductive lines **143** and vias **145** is formed on the dielectric layer **148**. In some embodiments, a seed layer (not shown) is first formed over the dielectric layer **148** and in openings through the dielectric layer **148**. In some embodiments, the seed layer is a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, the seed layer comprises a titanium layer and a copper layer over the titanium layer. The seed layer may be formed using, for example, PVD or the like. A photoresist is then formed and patterned on the seed layer. The photoresist may be formed by spin coating or the like and may be exposed to light for patterning. The pattern of the photoresist corresponds to the metallization pattern. The patterning forms openings through the photoresist to expose the seed layer. A conductive material is formed in the openings of the photoresist and on the exposed portions of the seed layer. The conductive material may be formed by plating, such as electroplating or electroless plating, or the like. The conductive material may comprise a metal, like copper, titanium, tungsten, aluminum, or the like. After forming the conductive material, the photoresist and portions of the seed layer on which the conductive material is not formed are removed. The photoresist may be removed by an acceptable ashing or stripping process, such as using an oxygen plasma or the like. Once the photoresist is removed, exposed portions of the seed layer are removed, for example using an acceptable etching process, such as a wet etching process or a dry etching process. The remaining portions of the seed layer and conductive material form the conductive lines **143** and vias **145**. The vias **145** are formed in openings through the dielectric layer **148** to make electrical connection to features

under the dielectric layer, such as to the through vias **119** and/or the die connectors **128**.

Turning to FIG. 8, additional dielectric layers (not individually labelled) and additional conductive features (not individually labelled) may be formed over dielectric layer **148** and conductive lines **143**, forming the redistribution structure **140**. The additional dielectric layers may be similar to the dielectric layer **148**, and the additional conductive features may be similar to the conductive lines **143** and vias **145**. The additional dielectric layers or additional conductive features may be formed similarly to the dielectric layer **148** or conductive lines **143** and vias **145**. For example, conductive features may be formed by forming openings in a dielectric layer of the redistribution structure **140** to expose underlying conductive features, forming a seed layer (not shown) over the dielectric layer and in the openings, forming a patterned photoresist (not shown) with a designed pattern over the seed layer, plating (e.g., electroplating or electroless plating) the conductive material in the designed pattern and over the seed layer, and removing the photoresist and portions of seed layer on which the conductive material is not formed. Other methods of forming the redistribution structure **140** are also possible and are fully intended to be included within the scope of the present disclosure.

The number of dielectric layers and the number of layers of the conductive features in the redistribution structure **140** of FIG. 8 are merely non-limiting examples. Other numbers of the dielectric layers and other numbers of layers of the conductive features are also possible and are fully intended to be included within the scope of the present disclosure.

FIG. 8 also illustrates under bump metallization (UBM) structures **147** formed over and electrically coupled to the redistribution structure **140**. In some embodiments, the UBM structures **147** are formed by first forming openings in the topmost dielectric layer of the redistribution structure **140** to expose conductive features (e.g., conductive lines or pads) of the redistribution structure **140**. After the openings are formed, the UBM structures **147** may be formed in electrical contact with the exposed conductive features. In an embodiment, the UBM structures **147** comprise three layers of conductive materials, such as a layer of titanium, a layer of copper, and a layer of nickel. However, there are many suitable arrangements of materials and layers, such as an arrangement of chrome/chrome-copper alloy/copper/gold, an arrangement of titanium/titanium tungsten/copper, or an arrangement of copper/nickel/gold, that are suitable for the formation of the UBM structures **147**. Any suitable materials or layers of material that may be used for the UBM structures **147** are fully intended to be included within the scope of the present disclosure.

The UBM structures **147** may be formed by forming a seed layer over the topmost dielectric layer (e.g., **142**) and along the interior of the openings in the topmost dielectric layer; forming a patterned mask layer (e.g., photoresist) over the seed layer; forming (e.g., by plating) the conductive material(s) in the openings of the patterned mask layer and over the seed layer; removing the mask layer and remove portions of the seed layer on which the conductive material(s) is not formed. Other methods for forming the UBM structures **147** are possible and are fully intended to be included within the scope of the present disclosure. Upper surfaces of the UBM structures **147** in FIG. 4 are illustrated to be planar merely as an example, the upper surfaces of the UBM structures **147** may not be planar. For example, portions (e.g., peripheral portions) of each UBM structure **147** may be formed over the topmost dielectric layer (e.g., **142**), and other portions (e.g., center portions) of each UBM

structure **147** may be formed conformally along sidewalls of the topmost dielectric layer exposed by a corresponding opening, as skilled artisans ready appreciate.

Next, in FIG. **9**, an electrical device **171** is attached to UBM structures **147** and connectors **155** are formed over UBM structures **147**, in accordance with some embodiments. The electrical device **171** may be a device, die, chip, or package, such as an integrated passive device (IPD) or the like. The electrical device **171** is electrically coupled to the redistribution structure **140** through the UBM structures **147** by conductive connectors **173**. The conductive connectors **173** may be, for example, solder connectors formed between the electrical device **171** and redistribution structure **140**. The conductive connectors **173** may comprise a same material (e.g., solder) as the connectors **155** (see below). In some embodiments, a flux material (not shown) may be deposited on the associated UBM structures **147** prior to placing the electrical device **171**. The electrical device **171** may be placed using, e.g., a pick and place process. In addition, an underfill material **175** may be formed in a gap between the electrical device **171** and the redistribution structure **140**. The electrical device **171** is optional, and may not be included in some embodiments.

Still referring to FIG. **9**, the connectors **155** may be solder balls, metal pillars, controlled collapse chip connection (C4) bumps, micro bumps, electroless nickel-electroless palladium-immersion gold technique (ENEPIG) formed bumps, combination thereof (e.g., a metal pillar having a solder ball attached thereof), or the like. The connectors **155** may include a conductive material such as solder, copper, aluminum, gold, nickel, silver, palladium, tin, the like, or a combination thereof. In some embodiments, the connectors **155** comprise a eutectic material and may comprise a solder bump or a solder ball, as examples. The solder material may be, for example, lead-based and lead-free solders, such as Pb—Sn compositions for lead-based solder; lead-free solders including InSb; tin, silver, and copper (SAC) compositions; and other eutectic materials that have a common melting point and form conductive solder connections in electrical applications. For lead-free solder, SAC solders of varying compositions may be used, such as SAC **105** (Sn 98.5%, Ag 1.0%, Cu 0.5%), SAC **305**, and SAC **405**, as examples. Lead-free connectors such as solder balls may be formed from SnCu compounds as well, without the use of silver (Ag). Alternatively, lead-free solder connectors may include tin and silver, Sn—Ag, without the use of copper. The connectors **155** may form a grid, such as a ball grid array (BGA). In some embodiments, a reflow process may be performed, giving the connectors **155** a shape of a partial sphere in some embodiments. In some cases, the reflow process may be performed on both the conductive connectors **173** and the connectors **155**. Alternatively, the connectors **155** may comprise other shapes. The connectors **155** may also comprise non-spherical conductive connectors, for example. In some embodiments, a flux material (not shown) may be formed over the associated UBM structures **147** prior to forming the connectors **155**.

In some embodiments, the connectors **155** comprise metal pillars (such as copper pillars), which may be formed by a sputtering, printing, electro plating, electroless plating, CVD, or the like, and may be formed with or without a solder material thereon. The metal pillars may be solder-free and have substantially vertical sidewalls or tapered sidewalls.

The structure shown in FIG. **9** is a single device package **1100** formed over the carrier **101**. One skilled in the art will appreciate that many packages (e.g., device package **1100**)

may be formed over a carrier substrate (e.g., carrier **101**) using similar processing steps as illustrated in FIGS. **1-9**. FIGS. **10-14** illustrate further processing of the semiconductor package **1100** of FIG. **9** according to some embodiments. The processing of FIGS. **10-14** is shown using two device packages (e.g., **1100A** and **1100B**) formed over a carrier **101**, with the understanding that more than two device packages may be formed over the carrier **101** in other embodiments.

FIG. **10** illustrates a structure comprising a device package **1100A** and a device package **1100B**, in accordance with some embodiments. The device package **1100A** and the device package **1100B** are formed in regions **100** and **200** over the carrier **101**, respectively. Each of the device packages **1100A** and **1100B** may be similar to the device package **1100** illustrated in FIG. **9**.

Turning to FIG. **11**, the structure shown in FIG. **10** is flipped over, and the external connectors **155** are attached to a tape **159** (e.g., a dicing tape) supported by a frame **157**, in accordance with some embodiments. Next, the carrier **101** is de-bonded from the composite layer **110** by a suitable process, such as etching, grinding, or mechanical peel off. In some embodiments in which an adhesive layer (e.g., an LTHC film) is formed between the carrier **101** and the composite layer **110**, the carrier **101** may be de-bonded by exposing the carrier **101** to a laser or UV light. The laser or UV light breaks the chemical bonds of the adhesive layer that binds to the carrier **101**, and the carrier **101** can then be detached. The adhesive layer may be removed by the carrier de-bonding process. After de-bonding the carrier **101**, a cleaning process may be performed on the composite layer **110** to remove any residue (e.g., from the adhesive layer).

Turning to FIG. **12**, after de-bonding the carrier **101**, openings **116** are formed in the composite layer **110** to expose the metallization pattern **112**, in accordance with some embodiments. In some embodiments, the openings **116** in the composite layer **110** may be formed using a suitable process such as a laser drilling process, an etching process, or the like. In some embodiments, the etching process is a plasma etching process. In some embodiments, a cleaning process is performed after forming the openings **116** in order to remove any residue (e.g., from a laser drilling process). Although not shown, solder paste may be formed in the openings **116** in preparation for attaching top packages (see FIG. **13**). The solder paste may be formed using a solder paste printing process or another suitable process.

Referring next to FIG. **13**, top packages **160** are attached to the device packages **1100** to form package structures **500**, in accordance with some embodiments. In FIG. **13**, example top packages **160A** and **160B** are shown attached to example device packages **1100A** and **1100B** to form example package structures **500A** and **500B**, respectively. In some embodiments, the package structures **500** may be package-on-package (PoP) or integrated fan-out (InFO-PoP) structures.

As illustrated in FIG. **13**, each of the top packages **160** (e.g., **160A**, **160B**) comprises a substrate **161** and one or more semiconductor dies **162** (e.g., memory dies) attached to an upper surface of the substrate **161**. In some embodiments, the substrate **161** includes silicon, gallium arsenide, silicon on insulator ("SOI"), the like, or a combination. In some embodiments, the substrate **161** is a multiple-layer circuit board. In some embodiments, the substrate **161** is formed from one or more materials such as bismaleimide triazine (BT) resin, FR-4 (a composite material composed of woven fiberglass cloth with an epoxy resin binder that is flame resistant), ceramic, glass, plastic, tape, film, or other supporting materials. The substrate **161** may include con-

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ductive features (e.g., conductive lines and vias, not shown) formed in or on the substrate **161**. As illustrated in FIG. **13**, the substrate **161** may have conductive pads **163** formed on the upper surface and a lower surface of the substrate **161**. The conductive pads **163** are electrically coupled to the conductive features of the substrate **161**, such as through vias or conductive lines. The one or more semiconductor dies **162** are electrically coupled to the conductive pads **163** by, e.g., bonding wires **167**. A molding material **165**, which may comprise an epoxy, an organic polymer, a polymer, encapsulant, or the like, is formed over the substrate **161** and around the semiconductor dies **162**. In some embodiments, the molding material **165** is continuous with the substrate **161**, as illustrated in FIG. **13**.

Still referring to FIG. **13**, the top packages **160** may be connected to device packages **1100** by conductive connectors **168** on the conductive pads **163**. The conductive connectors **168** make electrical connection between the metallization patterns **112** of the device packages **1100** and the conductive pads **163** of the top packages **160**. In some embodiments, a solder material **170** is deposited over the metallization pattern **112** exposed through the openings in the composite layer **110**. The conductive connectors **168** are attached to the solder material **170**. In some embodiments, the conductive connectors **168** comprise solder regions, conductive pillars (e.g., copper pillars with solder regions on at least end surfaces of the copper pillars), or the like. In some embodiments, a reflow process is performed to bond the solder material **170** and the conductive connectors **168**. After the reflow process, a baking process may be performed to remove moisture.

An underfill material **169** may then be formed in the gaps between the top packages **160** and the corresponding bottom packages **1100**. The underfill material **169** may be dispensed into gaps between the top packages **160** and the device packages **1100** using, e.g., a needle or a jetting dispenser. In some embodiments, a curing process may be performed to cure the underfill material **169**. Although not shown in FIG. **13**, the underfill material **169** may extend between or along sidewalls of the top packages **160**.

Next, in FIG. **14**, a singulation process is performed to separate the package structures **500** (e.g., **500A**, **500B**) into a plurality of individual package structures. After the singulation process is finished, a plurality of individual package structures, such as the package structure **500** illustrated in FIG. **14**, are formed. The singulation process may, for example, use a sawing process, a laser process, another suitable process, or a combination of processes.

In some cases, the use of a composite material for composite layer **110** (described previously with respect to FIG. **1**) can provide for improved rigidity of a package such as package structure **500**. The use of composite layer **110** in a package (e.g., package structure **500**) may reduce warping of that package, such as reducing warping of the device structure **1100** and/or reducing warping of the entire package structure **500**. For example, in some cases, a redistribution structure (e.g., redistribution structure **140**) may impart bending forces on the package which cause the package to warp or bend. The rigidity of the composite layer **110** can mitigate the warping due to these bending forces, and thus reduce the overall warping of the package. In some cases, the use of a composite layer such as composite layer **110** may reduce the bending distance of a warped package between about 0 μm and about 250 μm . In some cases, the use of a composite layer **110** may allow the package structure to have a bending distance of less than about 200 μm , such as less than about 80 μm or less than about 10 μm . In

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some cases, the use of a composite layer such as composite layer **110** may reduce the warping of a package between about 50% and about 100%. In some embodiments, the reduction in warping may be improved by disposing the composite layer **110** and the redistribution structure **140** on opposite sides of the die **120**.

Turning to FIGS. **15A-D**, illustrative close-up views of surfaces of the composite layer **110** are shown in accordance with some embodiments. FIG. **15A** illustrates a close-up view of the region labeled "A" in FIG. **14**, where the underfill material **169** has been deposited over the composite layer **110**. As shown in FIG. **15A**, the composite layer **110** has a pitted surface (also described above with respect to FIG. **1**). The pitted surface of the composite layer **110** can provide improved adhesion of the underfill material **169**, which can improve the overall rigidity of the package structure and reduce the chance of delamination. FIG. **15B** illustrates a close-up view of the region labeled "B" in FIG. **14**, which includes a sidewall of the composite layer **110**. As shown in FIG. **15B**, the sidewall of the composite layer **110** also has a pitted surface, which may improve adhesion of further materials deposited on the package structure **500** (e.g., molding compounds, encapsulants, or the like, which are not shown in the Figures). FIGS. **15C-D** illustrate close-up views of the region labeled "C" in FIG. **14**, which includes an opening in the composite layer **110** through which the solder material **170** extends (described previously with respect to FIG. **12**). FIG. **15C** shows the composite layer **110** with a tapered opening, and FIG. **15D** shows a composite layer with a substantially vertical opening, though the openings may have other shapes in other embodiments. As shown in FIGS. **15C-D**, the sidewalls of the openings may be pitted, and the solder material **170** may flow into the pits during deposition or during a reflow process. In this manner, the solder material **170** may have "bumps" corresponding to the pits in the sidewalls of the openings. In some cases, the pits can provide better adhesion of the solder material **170** to the composite layer **110**. Additionally, in some cases, the increased volume of the solder material **170** within the opening due to the presence of pits can reduce the resistance of the solder material **170** and thus improve electrical performance of the package structure **500**.

Referring next to FIG. **16**, a package structure **600** is shown, in accordance with some embodiments. The package structure **600** includes a top package **160**, which may be similar to top package **160** described previously (see FIG. **13**). The top package **160** is attached to a device package **1200** to form package structure **600**. The device package **1200** is similar to the device package **1100** described previously (see FIG. **9**), except that the dielectric layer **114** and the metallization pattern **112** are not formed over the composite layer **110** (see FIG. **3**). Thus, the through vias **119** and the molding material **130** are formed directly on the composite layer **110**. Portions of the molding material **130** may extend into pits of the pitted surface of the composite layer **110**. In some cases, the pitted surface of the composite layer **110** can provide improved adhesion of the molding material **130**. These and other variations of forming a package structure with a composite layer **110** are intended to be within the scope of this disclosure.

Embodiments may achieve advantages. By forming a package having conductive elements (e.g., solder material **170**) in a layer comprising a composite material (e.g., a polymer and filler), the rigidity of the package may be improved. In this manner, warping of the package may be reduced, and thus problems such as cracking or delamina-

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tion associated with warping may be reduced. Additionally, the composite material may form a layer having pitted surfaces, which can improve adhesion of other layers to the composite material, thus further improving the reliability and stability of the package.

In an embodiment, a method includes forming a composite material layer over a carrier, the composite material layer including particles of a filler material incorporated into a base material, forming a set of through vias over a first side of the composite material layer, attaching a die over the first side of the composite material layer, the die being spaced apart from the set of through vias, forming a molding material over the first side of the composite material layer, the molding material least laterally encapsulating the die and the through vias of the set of through vias, forming a redistribution structure over the die and the molding material, the redistribution structure electrically connected to the through vias, forming openings in a second side of the composite material layer opposite the first side, and forming conductive connectors in the openings, the conductive connectors electrically connected to the through vias. In an embodiment, the particles of the filler material have an average diameter that is between 0.5 μm and 30 μm . In an embodiment, the base material includes a polymer. In an embodiment, the filler material includes an oxide. In an embodiment, the method includes forming a dielectric layer over the composite material layer, wherein the material of the dielectric layer is different than the material of the composite material layer, and wherein the set of through vias are formed on the dielectric layer. In an embodiment, the method includes forming a metallization pattern on the composite material layer before forming the dielectric layer on the composite material layer. In an embodiment, forming the openings in the second side of the composite material layer includes a laser drilling process. In an embodiment, the openings in the second side of the composite material layer have pitted sidewalls. In an embodiment, the conductive connectors include a solder material, wherein the sidewalls of the conductive connectors within the composite material layer include a plurality of bumps extending laterally into the composite material layer. In an embodiment, the molding material physically contacts the first side of the composite material layer. In an embodiment, the die is physically attached to the first side of the composite material layer.

In an embodiment, a method includes forming a device package, wherein forming the device package includes forming a metallization pattern on a first surface of a composite layer, wherein the composite layer includes a composite material and wherein the first surface is pitted, forming a first dielectric layer over the composite layer and the metallization pattern, forming a conductive pillar over the first dielectric layer and electrically connected to the metallization pattern, placing a first semiconductor device on the first dielectric layer, wherein the first semiconductor device is adjacent to and separated from the conductive pillar, encapsulating the first semiconductor device and the conductive pillar with an encapsulant, and forming a redistribution structure over the encapsulant, forming openings in a second surface of the composite layer to expose the metallization pattern, and attaching a top package to the device package using conductive connectors, wherein the conductive connectors extend through the openings in the composite layer. In an embodiment, the composite layer has a Young's modulus between 10 GPa and 50 GPa. In an embodiment, the method includes depositing an underfill between the device package and the top package, the underfill surrounding the conductive connectors, wherein the

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underfill extends into the pits of the pitted top surface of the composite layer. In an embodiment, the device package has a bending distance less than 80 μm . In an embodiment, the method includes singulating the device package, wherein a sidewall surface of the device package comprises a plurality of pits. In an embodiment, the composite layer includes aluminum oxide incorporated into a polymer material.

In an embodiment, a semiconductor package includes a lower package including a die on a redistribution structure, the die electrically connected to a redistribution structure, a through via proximate the die and electrically connected to the redistribution structure, a molding material over the redistribution structure, the molding material being interposed between the die and the through via, and a composite layer over the die and the through via, the composite layer over a side of the die opposite the redistribution structure, and a top package comprising external connections, wherein the external connections are connected to the lower package through the composite layer. In an embodiment, an exposed sidewall of the composite layer has a pitted surface. In an embodiment, the semiconductor package includes an underfill material extending between the composite layer and the top package, wherein an interface between the underfill material and the composite layer is a surface including pitted regions.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A device comprising:

- a redistribution structure comprising a first metallization pattern and a first dielectric layer over the first metallization pattern;
- an integrated circuit die on the redistribution structure, wherein the integrated circuit die is electrically connected to the redistribution structure;
- an encapsulant on the redistribution structure and surrounding the integrated circuit die, wherein the encapsulant physically contacts sidewalls of the integrated circuit die;
- a conductive feature extending through the encapsulant, wherein the conductive feature is electrically connected to the redistribution structure;
- a polymer material extending over the integrated circuit die and extending on top surfaces of the conductive feature and the encapsulant, wherein the polymer material comprises a plurality of dielectric filler particles, wherein a top surface, interior sidewall surfaces, and exposed sidewall surfaces of the polymer material comprise a plurality of pits, and wherein a bottom surface of the polymer material comprises a plurality of pits; and
- a solder material extending through the polymer material, wherein the solder material is electrically connected to the conductive feature, wherein sidewall surfaces of the

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solder material comprise a plurality of protrusions that protrude into the polymer material at the interior sidewall surfaces.

2. The device of claim 1, wherein the top surface of the polymer material has a roughness in the range of 0.1 μm to 10 μm .

3. The device of claim 1, further comprising a package component attached to the solder material, wherein an underfill extends between the polymer material and the package component.

4. The device of claim 3, wherein the underfill fills the plurality of pits that are disposed in the top surface of the polymer material.

5. The device of claim 1, wherein the polymer material has a Young's modulus between 10 GPa and 50 GPa.

6. The device of claim 1 further comprising a metallization pattern between the conductive feature and the polymer material, wherein the metallization pattern is electrically connected to the conductive feature and to the solder material.

7. The device of claim 1, wherein an exposed sidewall of the polymer material and a sidewall of the encapsulant are coterminous.

8. The device of claim 1, wherein a pitted top surface of the polymer material is exposed.

9. The device of claim 1, wherein the polymer material physically contacts the top surfaces of the conductive feature and the encapsulant.

10. A device comprising:

a first package structure comprising:

a redistribution structure;

a plurality of through vias connected to the redistribution structure;

a composite layer over the through vias, wherein the composite layer comprises a filler material in a base material, wherein a top surface of the composite layer comprises a plurality of recesses, wherein a bottom surface of the composite layer comprises a plurality of recesses; and

a plurality of first connectors within the composite layer, wherein each through via of the plurality of through vias is connected to a respective first connector of the plurality of first connectors;

a second package structure attached to the first package structure, wherein the second package structure comprises a plurality of second connectors, wherein each

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second connector of the plurality of second connectors is connected to a respective first connector of the plurality of first connectors; and

an underfill material extending between the first package structure and the second package structure, wherein the underfill material fills recesses of the plurality of recesses in the top surface of the composite layer.

11. The device of claim 10, wherein the plurality of first connectors comprise a solder material.

12. The device of claim 10, wherein a portion of the top surface of the composite layer is exposed.

13. The device of claim 10, wherein a sidewall surface of the composite layer comprises a plurality of recesses.

14. The device of claim 13, wherein a first connector of the plurality of first connectors fills recesses of the plurality of recesses in the sidewall surface of the composite layer.

15. The device of claim 10, wherein the composite layer physically contacts the plurality of through vias.

16. The device of claim 10 further comprising a molding material encapsulating the plurality of through vias, wherein the composite layer physically contacts the molding material.

17. A method comprising:

attaching a die to a first side of a composite layer, wherein the composite layer comprises filler particles, wherein a surface of a first side of the composite layer is pitted; forming a through via on the first side of the composite layer;

depositing a molding material over the first side of the composite layer to encapsulate the die and the through via;

forming a redistribution structure extending over the molding material, the die, and the through via, wherein the redistribution structure is electrically connected to the die and the through via; and

forming a conductive material extending through the composite layer, wherein the conductive material is electrically connected to the through via.

18. The method of claim 17, wherein the surface of a second side of the composite layer is pitted.

19. The method of claim 17, wherein the conductive material comprises solder.

20. The method of claim 17 further comprising attaching a package component to the conductive material.

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